

FINANCIAL TRANSACTIONS AND LOANS DASHBOARD

Power BI Project Report

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Tool Used: Microsoft Power BI Desktop

1. Introduction

The Financial Transactions and Loans Dashboard is a data visualization project developed using Microsoft Power BI Desktop. The main purpose of this project is to analyse financial transactions, customer information, account types, and loan status.

The dashboard converts financial data into meaningful visualizations, making it easier to understand transaction patterns and financial performance. It includes KPI cards, charts, tables, filters, and interactive features to support effective data analysis.

2. Objectives of the Project

The main objectives of this project are:

- To analyse financial transactions and loan information.
- To display important financial indicators using KPI cards.
- To compare different types of accounts.
- To analyse transaction trends over time.
- To understand the distribution of loans by status.
- To compare transaction types and their values.
- To create an interactive dashboard using Power BI.
- To use drill-down and drill-up features for detailed analysis.

3. Tools Used

The following tools and features were used to develop the dashboard:

Microsoft Power BI Desktop: Used to create the dashboard and visualize financial data.

KPI Cards: Used to display important financial information, such as total customers, active accounts, active loans, and total transaction value.

Charts: Used to compare account types, transaction types, loan status, and transaction trends.

Slicers: Used to filter information based on account type, transaction type, and year.

Drill-Down and Drill-Up: Used to analyse transaction data at different levels of time.

Tooltip: Used to display additional transaction details when interacting with the report visuals.

4. Dashboard Overview

The Financial Transactions and Loans Dashboard provides a complete overview of financial activities through different visualizations.

The dashboard contains four KPI cards, an account type chart, a transaction trend line chart, a loan status doughnut chart, a transaction type bar chart, and a transaction summary table.

It also includes filters for account type, transaction type, and year. These features allow users to explore financial information according to their requirements.

5. Key Performance Indicators (KPIs)

The dashboard displays four important KPIs.

5.1 Total Customers

The Total Customers KPI displays **1,072 customers**. It helps users understand the total number of customers represented in the report.

5.2 Active Accounts

The Active Accounts KPI displays **1,297 active accounts**. It provides information about the number of active accounts in the dataset.

5.3 Active Loans

The Active Loans KPI displays **213 active loans**. It helps users understand the number of loans that are currently active.

5.4 Total Transaction Value

The Total Transaction Value KPI displays approximately **\$122.693 million**. It represents the total monetary value of transactions shown in the dashboard.

6. Visualizations Used in the Dashboard

6.1 Account by Account Type

A column chart is used to compare the number of accounts across five categories: Business, Youth, Savings, Payroll, and Checking.

The chart helps identify which account types have higher or lower account totals. In the displayed chart, Business accounts have the highest total.

This analysis helps users understand the distribution of accounts across different categories.

6.2 Transaction Over Time

A line chart is used to analyse transaction activity over different time periods.

The chart displays transaction information by month, quarter, and year. The monthly view shows a high transaction count in January, followed by a sharp decline in February.

After February, the transaction count fluctuates within a relatively narrow range.

This visualization helps identify changes and patterns in transaction activity over time.

6.3 Loans by Loan Status

A doughnut chart is used to display the distribution of loans according to their status.

The three categories are:

- **Active:** 213 loans (73.2%)
- **Paid Off:** 47 loans (16.15%)
- **Overdue:** 31 loans (10.65%)

The total number of loans displayed is 291.

The chart shows that active loans form the largest category. This visualization helps users understand the current status of the loan portfolio.

6.4 Transactions by Type

A horizontal bar chart is used to compare four types of transactions: Deposit, Transfer, Withdrawal, and Payment.

The displayed transaction counts are approximately:

- Deposit: 14.8K
- Transfer: 14.5K
- Withdrawal: 14.4K
- Payment: 4.9K

Deposit has the highest transaction count, while Payment has the lowest.

This chart helps users compare transaction activity across different categories.

6.5 Transaction Summary

The Transaction Summary table displays the total number of transactions and transaction value for each transaction type.

The table includes Deposit, Payment, Transfer, and Withdrawal.

The displayed overall totals are:

- **Total Transactions:** 48,510
- **Total Transaction Value:** \$122,693,050.81

This table provides a simple way to compare transaction counts and monetary values across categories.

7. Filters and Interactivity

The dashboard contains three main filters:

7.1 Account Type Filter

This filter allows users to select a specific account category, such as Business, Youth, Savings, Payroll, or Checking.

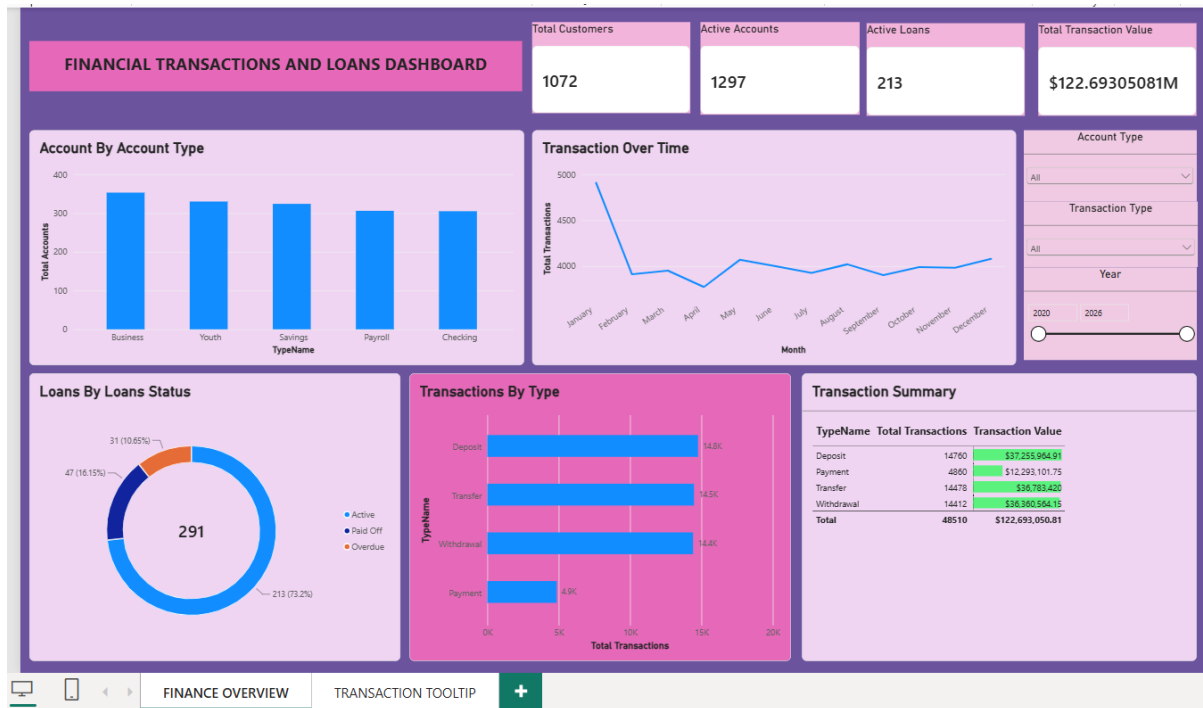
7.2 Transaction Type Filter

This filter allows users to view specific transaction categories, such as Deposit, Transfer, Withdrawal, or Payment.

7.3 Year Filter

The year filter allows users to select a particular year or date range for analysing transaction activity.

These filters make the dashboard interactive and help users focus on the information they need.



8. Drill-Down and Drill-Up Analysis

Drill-down and drill-up are interactive features in Power BI that allow users to analyse data at different levels of detail.

In this project, these features are used in the Transaction Over Time line chart to explore transaction activity by quarter, month, and year.

8.1 Quarter-Level Analysis

The quarter-level chart displays transaction information for four quarters: Qtr 1, Qtr 2, Qtr 3, and Qtr 4.

This view provides a summary of transaction activity across the year. It helps users compare the performance of different quarters and identify changes in transaction patterns.

8.2 Month-Level Analysis

The month-level chart displays transaction activity from January to December.

The chart shows that January has the highest transaction count. There is a significant decline in February, followed by moderate fluctuations during the remaining months.

This detailed view helps users understand monthly transaction patterns.

8.3 Year-Level Analysis

The year-level chart displays transaction information from 2020 to 2026.

The chart shows relatively high values from 2020 to 2023, followed by a sharp decline in 2024. The values remain low in 2025 and 2026.

This trend should be verified against the source dataset to ensure that the date filters and transaction measures are working correctly.

8.4 Drill-Down Function

Drill-down allows users to move from a higher-level view to a more detailed view.

For example, users can explore transaction information by year and then move to a more detailed time level.

Steps to use Drill Down:

1. Select the Transaction Over Time chart.
2. Click the Drill Down icon in the visual header.
3. Select a data point to explore the next level of detail.
4. Observe the changes in the chart.

8.5 Drill-Up Function

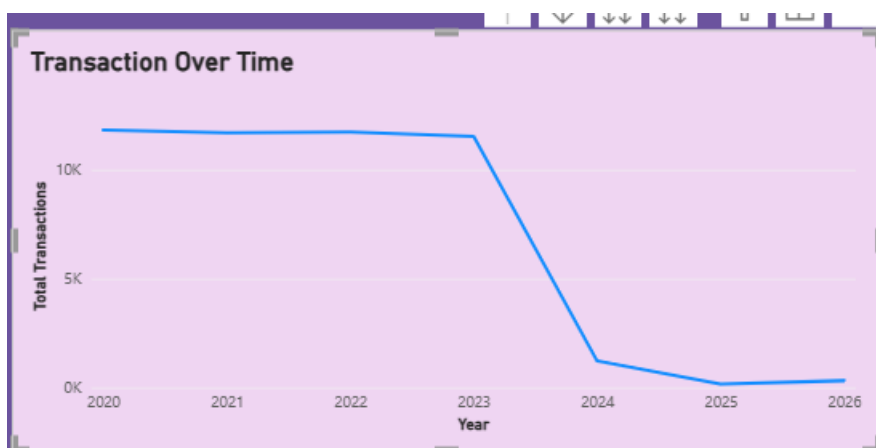
Drill-up allows users to return from a detailed view to a higher-level summary.

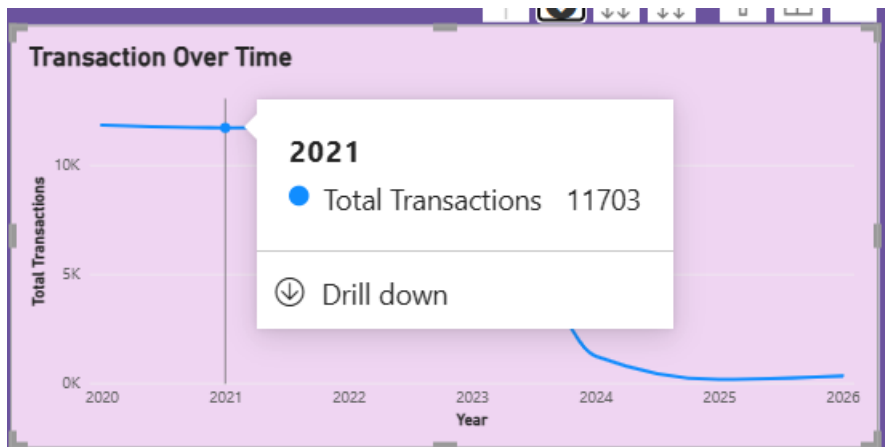
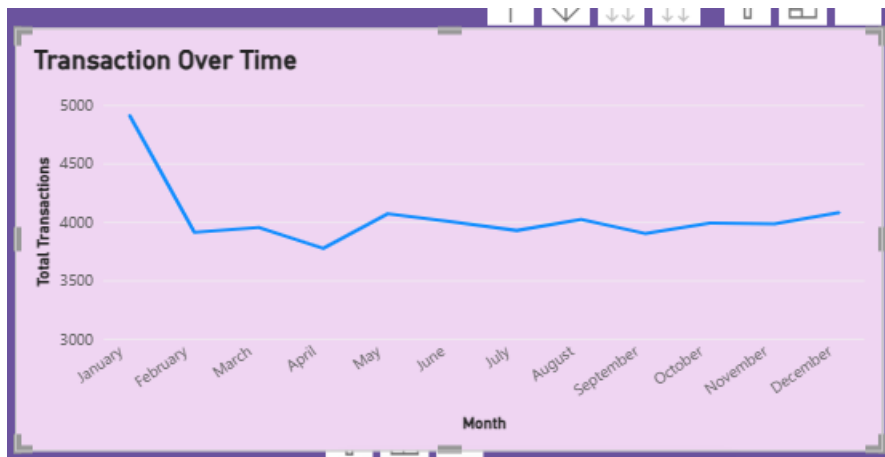
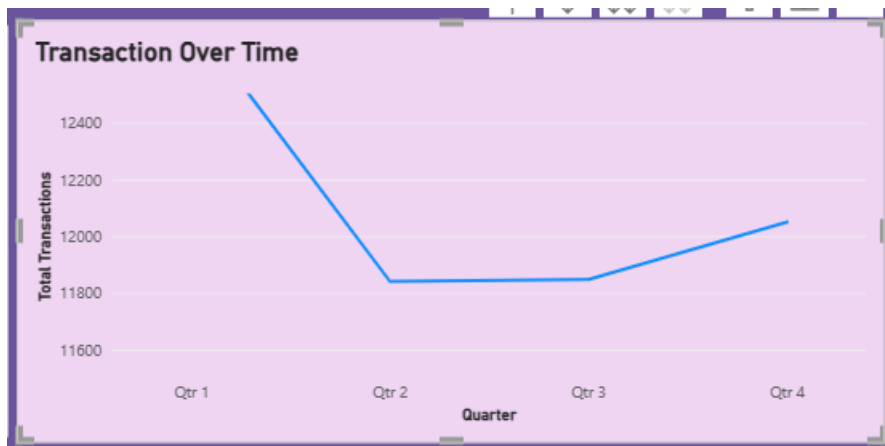
For example, users can move from a month-level view back to a quarter-level or year-level view, depending on the configured date hierarchy.

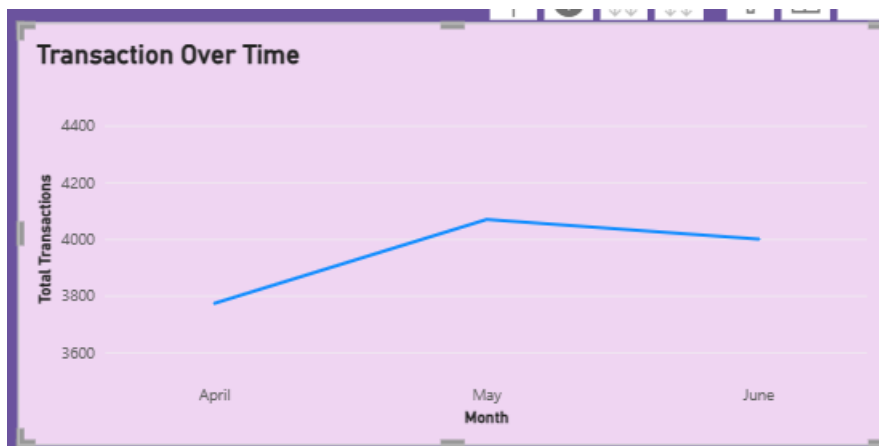
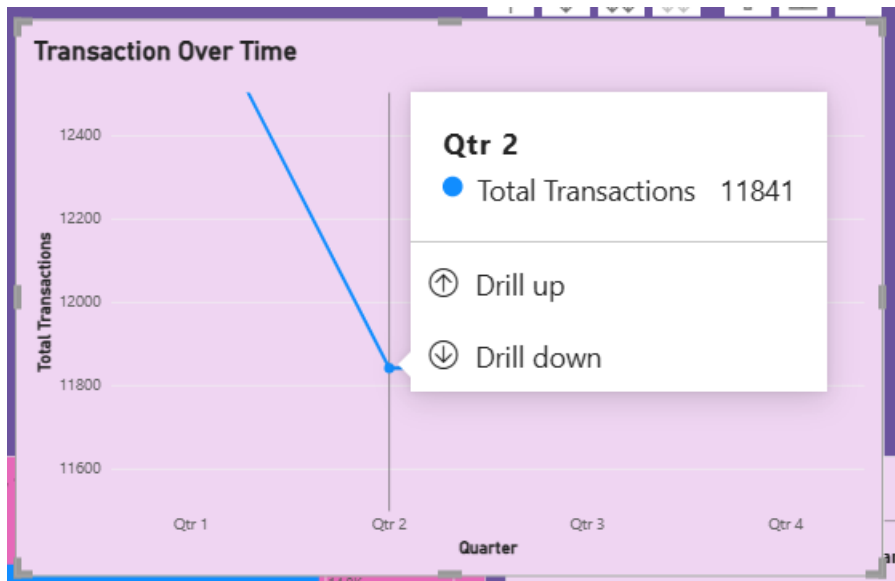
Steps to use Drill Up:

1. Select the Transaction Over Time chart.
2. Click the Drill Up icon in the visual header.
3. Observe the chart returning to the previous level.
4. Compare the broader transaction trend.

These features make it easier to explore financial data without creating separate charts for every time period.







9. Transaction Tooltip Page

The Transaction Tooltip page provides additional information when users hover over a data point in a visual.

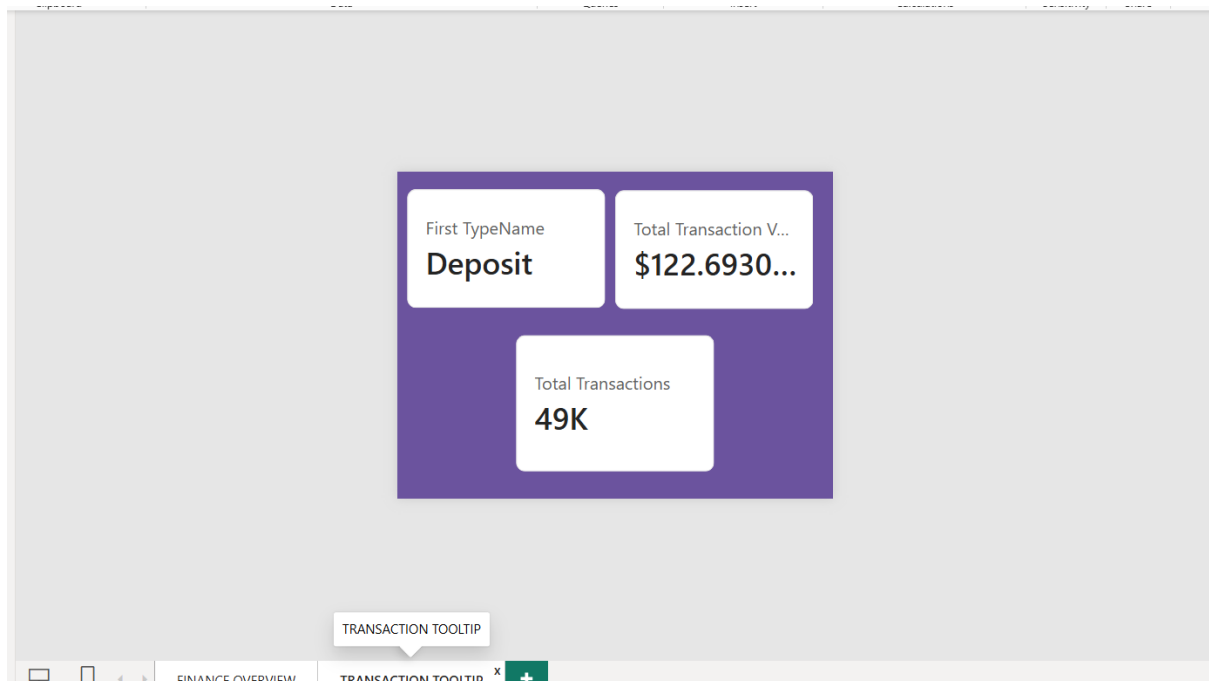
In this project, the tooltip page displays three important details:

- First TypeName
- Total Transaction Value
- Total Transactions

For example, when the transaction type is Deposit, the tooltip displays Deposit as the transaction type, along with the corresponding transaction value and transaction count.

The displayed transaction count is approximately 49K.

The tooltip feature makes the report more interactive and allows users to view additional information without navigating away from the main dashboard.



10. Key Findings

The following observations can be made from the dashboard:

1. Business accounts have the highest total among the displayed account types.
2. The dashboard displays 1,072 customers and 1,297 active accounts.
3. Active loans represent the largest category in the loan status chart.
4. Deposit has the highest transaction count among the displayed transaction types.
5. Payment has a comparatively lower transaction count.
6. January records the highest monthly transaction activity in the displayed line chart.
7. The transaction summary displays 48,510 transactions with a total value of \$122,693,050.81.
8. Drill-down and drill-up features allow transaction patterns to be analysed at different time levels.

11. Benefits of the Dashboard

The Financial Transactions and Loans Dashboard offers several benefits:

- **Easy Data Analysis:** Presents financial information through clear charts and tables.
- **Quick Decision-Making:** KPI cards provide a quick overview of financial performance.
- **Better Comparison:** Charts help compare account types and transaction categories.
- **Trend Identification:** Line charts help identify changes in transaction activity over time.
- **Interactive Reporting:** Slicers allow users to filter information according to their needs.
- **Detailed Analysis:** Drill-down and drill-up features provide different levels of detail.
- **Improved Understanding:** The tooltip page provides additional information about transactions.

12. Limitations and Data Validation

The accuracy of the dashboard depends on the quality of the source data and the correctness of the measures used in Power BI.

The transaction totals should be checked to ensure that the KPI cards and summary table display consistent values.

The sharp decline in the year-level chart after 2023 should also be verified against the original dataset.

Proper data validation helps ensure that the dashboard provides reliable information for analysis and decision-making.

13. Conclusion

The Financial Transactions and Loans Dashboard demonstrates how Microsoft Power BI can transform financial data into meaningful visual information.

The dashboard provides insights into customer numbers, account types, loan status, transaction categories, and transaction trends. KPI cards offer a quick summary, while charts and tables help users analyse financial information in detail.

The slicers, drill-down, drill-up, and tooltip features improve the interactivity of the report and make data exploration easier.

Overall, this project demonstrates the practical application of Power BI in financial data analysis and reporting. It helps users understand financial activities, identify transaction patterns, and support informed decision-making.